QUESTION-TO-SLIDE MAPPING

SECTION B - LECTURE 06

Thermal Comfort & Human Heat Balance

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	6.0 - Thermal Comfort (15 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Map the lecture to thermal-comfort questions by physiology: comfort significance, environmental variables, metabolism/heat production, heat gain/loss through conduction-convection-radiation-evaporation, thermal balance, vasomotor regulation, sweating/shivering, and objective/subjective variables. Comfort indices (ET/CET/EW/OT) and detailed “climatic situations” are not treated as fully covered because this deck does not teach them.

RESULT	Lecture 06 is a major repeated-theory deck. It directly supports the recurring families on human heat exchange, thermal comfort variables, heat production, thermal balance, vasomotor adjustment and subjective variables. Its main gaps are comfort indices and the separate four-situation heat-loss analysis.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

The lecture explains why thermal comfort matters to health and efficiency, identifies the four environmental variables controlling thermal response, defines metabolism and activity heat production, describes body heat gain and loss by conduction, convection, radiation and evaporation, states a course thermal-balance equation, and explains physiological regulation by vasomotor adjustment, sweating and shivering. It closes by separating objective climatic variables from subjective individual variables. This content aligns with a large number of repeated questions across the supplied papers.

15	26	18	1	7	310
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
6-9	Human heat gain/loss + four heat-transfer paths	2016, 2018, 2020 (+ related repeats)	Highest-value explanation + sketch block.
10-12	Thermal balance + vasomotor / sweating / shivering	2016, 2021, 2022, 2023, 2024	Very high recent repeat value.
13-14	Objective vs subjective thermal-comfort variables	2018, 2022, 2023, 2024	Short-to-medium repeated questions.
2-5	Thermal comfort significance + metabolism + heat production	2016, 2018, 2022, 2023, 2024	Foundation for long answers.
Deck gap	ET/CET/EW/OT and detailed climatic-situation cases	2016-2023 separate families	Must use another lecture/source.

Priority note — Memorize one master flow: Metabolism -> heat gain/loss paths -> thermal balance -> vasomotor response -> sweating/shivering. Add objective vs subjective variables at the end. This structure answers most of the repeated long questions from this lecture.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q5(b)	20	1	Explain the human body heat exchange process.	Exact	4-9	Metabolism, heat gain and heat loss through conduction, convection, radiation and evaporation are directly explained with a body-exchange sketch.
2016	Q6(a)	15	1	What are the factors of Climate Comfort? Explain with examples.	Exact	3, 13-14	Air temperature, humidity, radiation and air movement are given as objective variables; individual variables are also listed.
2016	Q6(b)	10	1	What are the thermal comfort ideas?	Exact	2-3	The lecture directly explains comfort, stress, health/efficiency and the environmental variables affecting response.
2016	Q7(a)	20	1	Specify the heat exchange process for a built environment with necessary sketch and example.	Exact	6-9	Direct heat-exchange mechanisms and sketch support.
2016	Q7(b)	8	1	What is the thermal balance equation?	Exact	10	The course balance equation is explicitly printed.
2018	Q4(a)	20	3	Describe the processes of heat loss in various climatic situations.	Partial	7-9	The basic heat-loss mechanisms are covered, but the lecture does not organize them into the specific climatic situations demanded by the question.
2018	Q4(b)	15	3	Can you explain what and how happens in the process of human body heat exchange with necessary sketches?	Exact	4-9	Direct match.
2018	Q7(a)	20	4	What is thermal comfort? How does the human body produce and release heat to the environment?	Exact	2-9	Comfort significance, metabolism/heat production and heat-release mechanisms are all present.
2018	Q7(b)	5	4	Outline the various subjective variables by which thermal preferences are influenced.	Exact	14	Direct list match.
2018	Q7(c)	10	4	What is “effective temperature”? Discuss its limitations.	Partial	3, 13-14	The deck supplies thermal variables but does not define Effective Temperature or its limitations.
2020	Q1(a)	15	5	Describe human body heat exchange with environment through conduction, convection, radiation and evaporation.	Exact	6-9	Direct four-process match.
2021	Q4(a)	13	6	Write down the thermal balance equation of the human body. What happens if the equation is unbalanced?	Exact	10-12	Balance equation and the physiological warm/cold responses are directly developed.
2021	Q5(b)	12	7	Discuss the process of heat loss in various climatic situations.	Partial	7-9	Mechanisms are direct; climatic-situation variants require another source/lecture.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2021	Q5(c)	10	7	Discuss the designer's responsibilities in ensuring thermal comfort in the built environment.	Partial	2-3	The deck establishes why designers must ensure comfort, but does not provide a full responsibility/design-action list.
2022	Q3(d)	10	8	Describe the human body's mechanism of heat production and outline subjective variables influencing thermal comfort.	Exact	4-5, 14	Heat production and subjective variables are both directly covered.
2022	Q5(b)	12	8	Discuss the process of heat loss in various climatic situations.	Partial	7-9	Same coverage limitation as 2021.
2022	Q6(d)	6	9	Explain vasomotor adjustment and how the autonomic nervous system guides dilation/constriction under ambient temperatures.	Exact	10-12	Warm and cold vasomotor responses are directly explained.
2023	Q4(b)	8	10	Describe the objective variables affecting thermal comfort.	Exact	13	Radiation, air temperature, humidity and air movement are directly listed.
2023	Q4(c)	10	10	Describe the significance of thermal comfort for the human body.	Exact	2-3	Health, efficiency, physical/mental well-being and climatic stress are directly discussed.
2023	Q4(d)	12	10	Write the thermal-balance equation and describe regulating mechanisms when the equation is unbalanced.	Exact	10-12	Direct equation + vasomotor + sweating/shivering match.
2023	Q5(a)	8	11	Briefly explain vasomotor adjustment and the autonomic nervous system.	Exact	10-12	Direct repeated match.
2023	Q6(c)	14	11	Discuss the process of heat loss or gain in various climatic conditions.	Partial	6-9	General gain/loss mechanisms are present; the separate climatic-condition cases are not.
2024	Q3(c)	12	13	What is thermal comfort? How does the human body maintain thermal balance?	Exact	2-3, 10-12	Definition/significance plus thermal-balance regulation are directly supported.
2024	Q3(d)	5	13	Outline the various subjective variables by which thermal preferences are influenced.	Exact	14	Direct list match.
2024	Q6(a)	10	14	Explain vasomotor adjustment as part of human thermal regulation and analyse its relationship to comfortable indoor design.	Direct	10-12	Physiological regulation is exact; the design implication must be inferred/applied by the student.
2024	Q6(b)	10	14	Discuss a designer's responsibilities for comfortable indoor environments and domains of essential knowledge.	Partial	2-3	The lecture establishes the need for comfort knowledge but does not teach the full responsibility/knowledge-domain framework.

Coverage key	Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

Thermal-comfort questions vary in wording but usually test the same physiological sequence. A strong answer begins with why comfort matters, then explains heat production, four exchange paths, balance and regulation. Shorter questions isolate objective/subjective variables or vasomotor adjustment.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Human body heat exchange: conduction / convection / radiation / evaporation	2016, 2018, 2020 (+ related)	6-9	Classic long-answer family with a must-draw sketch.
S	Thermal balance equation + physiological regulation	2016, 2021, 2023, 2024	10-12	Repeated direct equation/mechanism family.
S	Thermal comfort factors / significance / objective variables	2016, 2018, 2023, 2024	2-3, 13	Stable concept family.
A	Vasomotor adjustment	2022, 2023, 2024	10-12	Three recent consecutive papers.
A	Subjective variables	2018, 2022, 2024 (+ related)	14	Easy short-note marks.
B	Heat loss in various climatic situations	2018, 2021, 2022, 2023	7-9 partial	Highly repeated, but another source is needed for the specific situations.
B	Effective temperature / comfort indices	2016-2023	Not taught here	Do not attempt from this deck alone.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
“human body heat exchange”	4-9	Start metabolism -> heat gain -> convection/radiation/evaporation/conduction; draw body sketch.
“thermal balance equation / unbalanced”	10-12	Use course equation; >0 -> more skin blood flow / heat loss; <0 -> reduced skin flow; then sweating or shivering if needed.
“vasomotor adjustment”	10-12	Warm: vasodilation; Cold: vasoconstriction; if insufficient -> sweating / shivering.
“objective variables”	13	Radiation, air temperature, humidity, air movement.
“subjective variables”	14	Clothing, acclimatization, age/sex, body shape/fat, health, food/drink, skin colour.
“effective temperature / ET-CET-EW-OT”	Other comfort-index lecture	This deck does not define or derive the indices.
“various climatic situations”	Other heat-loss lecture/note	Use Slides 7-9 only for mechanism; add the specific cases from the correct source.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
2-3	Thermal comfort significance + four environmental response variables	2016 Q6(a,b); 2018 Q7(a); 2023 Q4(c); 2024 Q3(c)	Prepare definition + why it matters.
4-5	Metabolism and rate of heat production	2018 Q7(a); 2022 Q3(d)	Know basal vs muscular metabolism and the 20%/80% lecture statement.
6-9	Heat gain / heat loss / exchange mechanisms	2016 Q5(b), Q7(a); 2018 Q4(b); 2020 Q1(a); climatic-situation questions partial	Draw one labelled body heat-exchange diagram.
10-12	Thermal balance + vasomotor regulation + sweating/shivering	2016 Q7(b); 2021 Q4(a); 2022 Q6(d); 2023 Q4(d), Q5(a); 2024 Q3(c), Q6(a)	Highest priority recent physiology block.
13	Objective variables	2023 Q4(b); supports comfort-factor questions	Memorize R-T-H-A.
14	Subjective variables	2018 Q7(b); 2022 Q3(d); 2024 Q3(d)	Memorize as a compact list.
15	Closing	None	Skip.

6. Coverage Gaps & Diagnostic Exclusions

Two high-frequency question families sit near this lecture but are not fully taught by it: comfort indices and heat-loss/gain “in various climatic situations.” The report keeps those mappings Partial instead of pretending the slide deck contains the missing material.

Gap	Affected questions	Action
ET / CET / EW / OT are not defined	2016 Q6(c); 2017 CET short note; 2018 Q7(c); 2021-2023 comfort-index questions	Use the dedicated comfort-index lecture/Section A material.
Specific heat-loss climatic situations are not organized here	2018 Q4(a); 2021/2022 Q5(b); 2023 Q6(c)	Use the lecture/note that explicitly teaches the separate climatic cases.
Designer responsibilities / essential knowledge are only introduced conceptually	2021 Q5(c); 2024 Q6(b)	Use the dedicated designer-responsibility lecture.
The course balance equation uses its own notation	Balance questions	Reproduce the notation taught in class; do not silently replace it with a different external convention.

Mapping discipline The report preserves the lecture's own physiology and equation wording. It does not import comfort-index formulas or additional thermal models into an Exact match. Course-specific statements such as the 20% consumed / 80% surplus heat split are treated as slide content for exam consistency.

7. Recommended Study Plan

Study this lecture as one connected physiological story. It is much easier to remember when every question is placed on the same chain from metabolism to regulation.

Step	Study block	Slides	Practice immediately
1	Human heat exchange	4-9	2016 Q5(b), 2018 Q4(b), 2020 Q1(a).
2	Thermal balance + vasomotor regulation	10-12	2021 Q4(a), 2022 Q6(d), 2023 Q4(d), 2024 Q6(a).
3	Comfort significance + objective variables	2-3, 13	2016 Q6(a,b), 2023 Q4(b,c), 2024 Q3(c).
4	Subjective variables	14	2018 Q7(b), 2024 Q3(d).
5	Gap check: climatic situations + comfort indices	7-9 / other lecture	Switch to the dedicated source before attempting those repeated questions.

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	Human body heat-exchange sketch	Label convection, radiation, evaporation and conduction; connect to metabolism.
A	Thermal-balance flow	Equation -> too warm response -> too cold response -> sweating/shivering.
A	Objective vs subjective table	Environment on left; individual/person factors on right.
A	Metabolism note	Basal vs muscular + rate of heat production link to activity.

Bottom line	Lecture 06 is a core thermal-comfort scoring deck. Master heat exchange and regulation first, then objective/subjective variables. Do not use it alone for ET/CET/EW/OT or the separate climatic-situation heat-loss question.